

LAB ACTION PLAN FOR WEEK 11

Jenkins-CI/CD

1. CI-Continuous Integration using Webhooks .
2. Sending E-mail Notification on Build Failure or success
3. Upload the screenshots for the tasks

Lab

Setting Up Jenkins CI-----using GitHub Webhook with Jenkins

Step 1: Configure Webhook in GitHub

1. Go to your GitHub repository.
2. Navigate to Settings → **Webhooks**.
3. Click **"Add webhook"**.
4. In the Payload URL field:
 - Enter the Jenkins webhook URL in the format:
http://<jenkins-server-url>/github-webhook/

Note: If Jenkins is running on localhost, GitHub cannot access it directly.

Use [ngrok](#) to expose your local Jenkins to the internet:

- ngrok.exe http <Jenkins local host:8080>
 - Use the generated ngrok URL, e.g.:
 - http://abc123.ngrok.io/**github-webhook/**
5. Set Content type to:
application/json
 6. Under "Which events would you like to trigger this webhook?", select:
 - Just the push event
 7. Click "Add webhook" to save.

Step 2: Configure Jenkins to Accept GitHub Webhooks

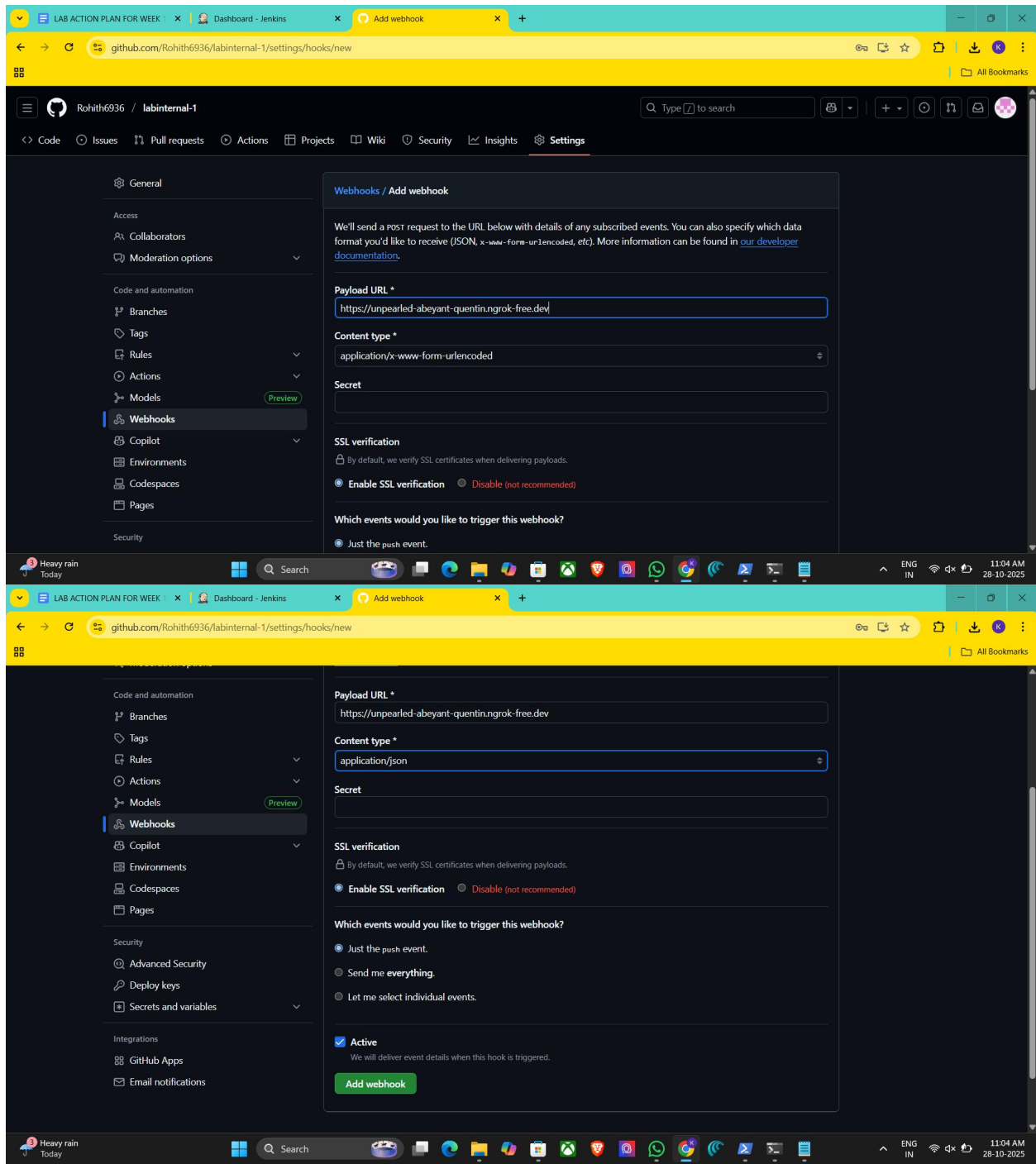
1. Open Jenkins Dashboard.
2. Select the job (freestyle or pipeline) you've already created.
3. Click Configure.
4. Scroll down to the Build Triggers section.
5. Check the box: ☒ GitHub hook trigger for GITScm polling
6. Click Save.

Step 3: Test the Setup

1. Make any code update in your local repo and push it to GitHub.
2. Once pushed, GitHub will trigger the webhook.
3. Jenkins will automatically detect the change and start the build pipeline.

outcome

- You've successfully connected GitHub and Jenkins using webhooks.
- Every time you push code to GitHub, Jenkins will automatically start building your project without manual intervention.



LAB ACTION PLAN FOR WEEK

mvnweb_build Config - Jenkins

Webhooks - Settings - Rohith60

localhost:8080/job/mvnweb_build/configure

All Bookmarks

Jenkins

mvnweb_build / Configuration

Configure

General

Source Code Management

Triggers

Environment

Build Steps

Post-build Actions

(Auto)

Additional Behaviours

Add

Triggers

Set up automated actions that start your build based on specific events, like code changes or scheduled times.

☐ Trigger builds remotely (e.g., from scripts)

☐ Build after other projects are built

☐ Build periodically

☒ GitHub hook trigger for GITScm polling

☐ Poll SCM

Environment

Configure settings and variables that define the context in which your build runs, like credentials, paths, and global parameters.

Saved

Save

Apply

LAB ACTION PLAN FOR WEEK

mvnweb_build Config - Jenkins

Webhooks - Settings - Rohith60

localhost:8080/job/mvnweb_build/configure

All Bookmarks

Jenkins

mvnweb_build / Configuration

Configure

General

Source Code Management

Triggers

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Set up automated actions that start your build based on specific events, like code changes or scheduled times.

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☐ Build after other projects are built

☐ Build periodically

☒ GitHub hook trigger for GITScm polling

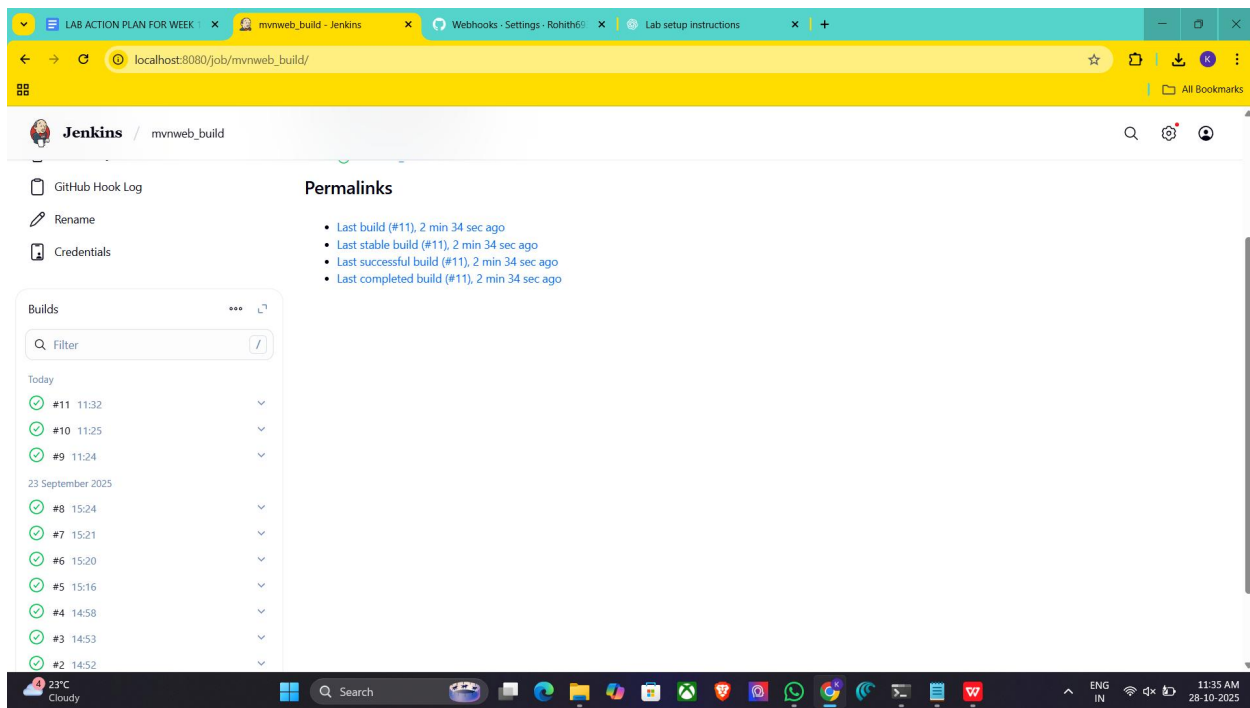
☐ Poll SCM

Environment

Configure settings and variables that define the context in which your build runs, like credentials, paths, and global parameters.

Save

Apply



Set-uping the ngrok

How to Install and Use ngrok

Step 1. Download ngrok

<https://ngrok.com/download>

Download and extract it for your OS (Windows, macOS, or Linux).

Step 2. Connect Your ngrok Account (optional but useful)

After you sign up (free), ngrok gives you an auth token.

CREATE AUTHENTICATOR [<https://dashboard.ngrok.com/get-started/your-authtoken>]

Run this command (replace <your_token> with yours):

ngrok config add-authtoken <your_token>

This ensures stable sessions and more control.

Step 3. Start a Tunnel for Jenkins

Assuming Jenkins runs locally on port 8085:

ngrok http 8085

You'll see output like:

Session Status online

Forwarding <https://1234abcd.ngrok.io> -> <http://localhost:8080>

Copy the HTTPS URL (<https://1234abcd.ngrok.io>) — this is your public Jenkins URL for webhooks.

Step 4. Use it in GitHub Webhook

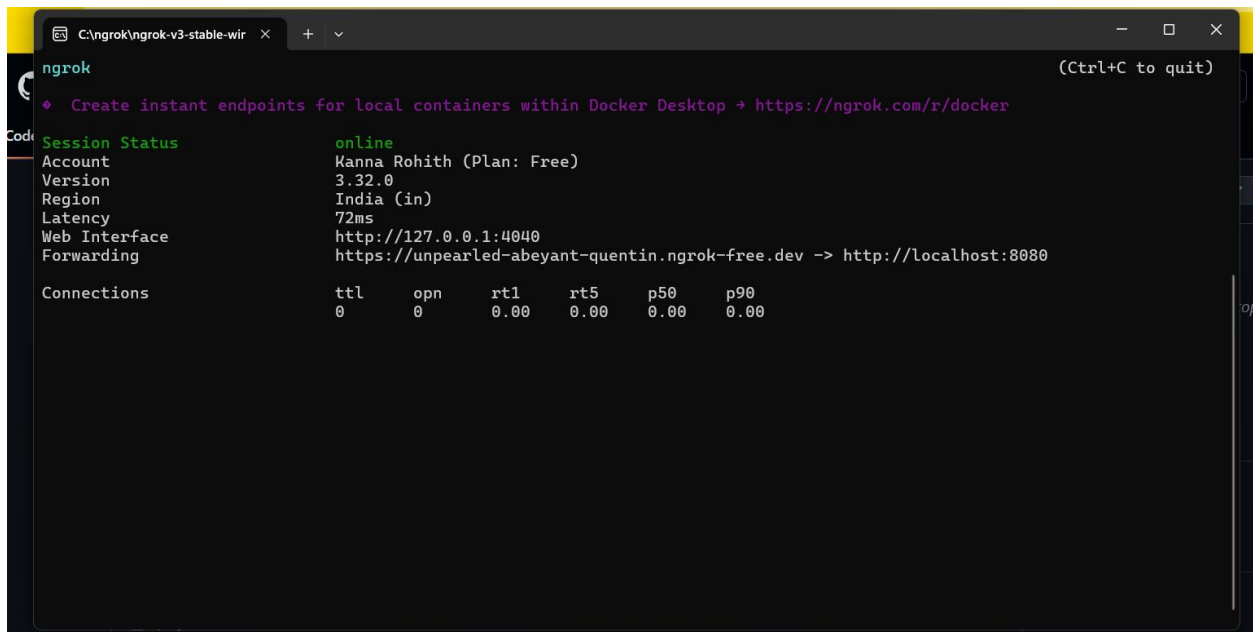
In your GitHub repo → Settings → Webhooks:

- Payload URL: *[paste the url generated by ngrok]*

<https://1234abcd.ngrok.io/github-webhook/> [please include this – remaining all default]

Now, whenever you push code, GitHub sends an event to that URL, which ngrok forwards to your local Jenkins.

The screenshot displays a Windows 10 desktop environment. The top portion shows the Microsoft Store application window with the ngrok product page. The page includes the ngrok logo, a description of the tool as a globally distributed reverse proxy, and a 'Screenshots' section. One screenshot shows a terminal window with the command `ngrok http --domain=www.example.com 8080` and a browser window displaying 'This URL is now available.' Below the Microsoft Store window, the Windows taskbar is visible, showing several open browser tabs: 'LAB ACTION PLAN FOR WEEK', 'Dashboard - Jenkins', 'ChatGPT', and 'Setup - ngrok'. The active tab is 'Setup - ngrok', which displays the ngrok dashboard at dashboard.ngrok.com/get-started/setup/windows. The dashboard has a dark theme and a sidebar menu on the left with categories like 'Getting Started', 'Universal Gateway', 'Traffic Observability', 'Billing & Usage', 'Early Access', 'Support', and 'Documentation'. The main content area is titled 'Connect' and 'Get an endpoint online.' It features a 'Windows' agent selection card, an 'Installation' section with links to 'Microsoft Store', 'WinGet via Microsoft Store', 'Scoop', and 'Download', and a 'Deploy your app online' section. A command is provided to add an auth token to the `ngrok.yml` configuration file: `ngrok config add-authtoken 34gF5p0ASn10FY4aN7jcvcnBVX1_5YLVNNy18XAGZGnhZwkJf`. The desktop taskbar at the bottom shows the system clock as 10:23 AM on 28-10-2025.



The screenshot shows the ngrok application window. At the top, there's a title bar with the file path 'C:\ngrok\ngrok-v3-stable-win'. Below the title bar, the ngrok logo is on the left, and '(Ctrl+C to quit)' is on the right. A purple link 'Create instant endpoints for local containers within Docker Desktop → https://ngrok.com/r/docker' is displayed. The main content area shows 'Session Status' with the following details: Account: Kanna Rohith (Plan: Free), Version: 3.32.0, Region: India (in), Latency: 72ms, Web Interface: http://127.0.0.1:4040, and Forwarding: https://unpearled-abeyant-quentin.ngrok-free.dev -> http://localhost:8080. Below this, a table shows connection statistics:

Connections	t1	opn	rt1	rt5	p50	p90
	0	0	0.00	0.00	0.00	0.00

Setting Up Jenkins Email Notification Setup (Using Gmail with App Password)

Creation of app password

1. Gmail: Enable App Password (for 2-Step Verification)

i. Go to: <https://myaccount.google.com>

ii. Enable 2-Step Verification

- Navigate to:
 - Security → 2-Step Verification
 - Turn it **ON**
 - Complete the OTP verification process (via phone/email)

iii. Generate App Password for Jenkins

- Go to:
 - Security → App passwords
- Select:
 - **App:** Other (Custom name)
 - **Name:** Jenkins-Demo
- Click **Generate**
- Copy the **16-digit app password**
 - Save it in a secure location (e.g., Notepad)

2. Jenkins Plugin Installation

i. Open Jenkins Dashboard

ii. Navigate to:

- Manage Jenkins → Manage Plugins

iii. Install Plugin:

- Search for and install:
 - Email Extension Plugin
-

3. Configure Jenkins Global Email Settings

i. Go to:

- Manage Jenkins → Configure System

A. E-mail Notification Section

Field	Value
SMTP Server	smtp.gmail.com
Use SMTP Auth	✔ Enabled
User Name	Your Gmail ID (e.g., archanareddykmit@gmail.com)
Password	Paste the 16-digit App Password
Use SSL	✔ Enabled
SMTP Port	465

Reply-To Address Your Gmail ID (same as above)

➤ Test Configuration

- Click: Test configuration by sending test e-mail
- Provide a valid email address to receive a test mail
- ✔ Should receive email from Jenkins

B. Extended E-mail Notification Section

Field	Value
SMTP Server	smtp.gmail.com
SMTP Port	465
Use SSL	✔ Enabled
Credentials	Add Gmail ID and App Password as Jenkins credentials
Default Content Type	text/html or leave default
Default Recipients	Leave empty or provide default emails
Triggers	Select as per needs (e.g., Failure)

4. Configure Email Notifications for a Jenkins Job

i. Go to:

- Jenkins → Select a Job → Configure
-

ii. In the Post-build Actions section:

- Click: Add post-build action → **Editable Email Notification**

A. Fill in the fields:

Field	Value
Project Recipient List	Add recipient email addresses (comma-separated)
Content Type	Default (text/plain) or text/html
Triggers	Select events (e.g., Failure, Success, etc.)
Attachments	(Optional) Add logs, reports, etc.

iii. Click Save

Now your Jenkins job is set up to send email notifications based on the build status!

- ---

Takeaway :

Students learned how to integrate Jenkins with GitHub using webhooks to automate build triggers and configure email notifications to monitor build success or failure effectively.

LAB ACTION PLAN FOR WEEK X mvnweb_build - Jenkins Webhooks - Settings - Rohith6 Lab setup instructions Security

myaccount.google.com/security

Google Account Search Google Account

Home Personal info Data and privacy **Security** People and sharing Payments and subscriptions About

New sign-in on Windows 10 Oct · Telangana, India
[Review security activity \(8\)](#)

How you sign in to Google
Make sure that you can always access your Google Account by keeping this information up to date

2-Step Verification	On since 7 Oct	>
Passkeys and security keys	3 passkeys	>
Password	Last changed 10 Oct	>
Skip password when possible	On	>
Google prompt	1 device	>
2-Step Verification phones	084999 23053	>
Recovery phone	084999 23053	>

Privacy Terms Help About
https://myaccount.google.com/signinoptions/passkeys?utm_source=google-account&utm...

Rainy days ahead 23°C

Search

ENG IN 11:50 AM 28-10-2025

LAB ACTION PLAN FOR WEEK X mvnweb_build - Jenkins Webhooks - Settings - Rohith6 Lab setup instructions App passwords

myaccount.google.com/apppasswords?continue=https://myaccount.google.com/security&rapt=AEjHL4OMhk-beJSj_s1Z8c7RQIEyz2FOOHTKdJ9WG2G0488N-JWNhbgZcpSFFTEWM...

Google Account

← App passwords

App passwords help you sign in to your Google Account on older apps and services that don't support modern security standards.

App passwords are less secure than using up-to-date apps and services that use modern security standards. Before you create an app password, you should check to see if your app needs this in order to sign in.
[Learn more](#)

Your app passwords

SE_LAB	Created on 11:55	🗑️
--------	------------------	----

To create a new app-specific password, type a name for it below...

App name

Create

2 cm of rain Today

Search

ENG IN 11:56 AM 28-10-2025

LAB ACTION PLAN FOR WEEK XInstalled plugins - Plugins - JerWebhooks - Settings - Rohith6Lab setup instructionsApp passwords

localhost:8080/manage/pluginManager/installed

☆📄📌🔍⋮

All Bookmarks

Jenkins / Manage Jenkins / Plugins

🔍 email

Updates 37

Available plugins

Installed plugins

Advanced settings

Name ↓	Health	Enabled
Email Extension 1925.v1598902b_58dd This plugin is a replacement for Jenkins's email publisher. It allows to configure every aspect of email notifications: when an email is sent, who should receive it and what the email says Report an issue with this plugin	100	☒
Mailer Plugin 522.va_995fa_cfb_8b_d This plugin allows you to configure email notifications for build results. Report an issue with this plugin	100	☒

REST API Jenkins 2.516.3

2 cm of rain Wednesday

🔍 Search

📄📌🔍⋮

All Bookmarks

LAB ACTION PLAN FOR WEEK XAvailable plugins - Plugins - JerWebhooks - Settings - Rohith6Lab setup instructionsApp passwords

localhost:8080/manage/pluginManager/available

☆📄📌🔍⋮

All Bookmarks

Jenkins / Manage Jenkins / Plugins

🔍 Search available plugins

📄 Install📄🔄

Updates 37

Available plugins

Installed plugins

Advanced settings

Install	Name ↓	Released	Health
☐	Pipeline Graph Analysis 245.v8f03631a_b_21 Library plugins (for use by other plugins) Provides a REST API to access pipeline and pipeline run data.	1 mo 18 days ago	97
☐	PAM Authentication 1.12 Security Adds Unix Pluggable Authentication Module (PAM) support to Jenkins	7 mo 27 days ago	95
☐	JavaMail API 1.6.2-11 Library plugins (for use by other plugins) This plugin provides the JavaMail API for other plugins.	8 mo 7 days ago	96
☐	Command Agent Launcher 123.v37cfd92ef67 Agent Management Allows agents to be launched using a specified command.	6 mo 20 days ago	100
☐	Oracle Java SE Development Kit Installer 83.v417146707a_3d Allows the Oracle Java SE Development Kit (JDK) to be installed via download from Oracle's website.	9 mo 9 days ago	100
☐	SSH server 3.374.v19b_d59ce6610 Adds SSH server functionality to Jenkins, exposing CLI commands through it.	2 mo 18 days ago	97

LAB ACTION PLAN FOR WEEK x localhost:8080/manage/config Webhooks - Settings - Rohith6 Lab setup instructions App passwords

localhost:8080/manage/configure

Jenkins / Manage Jenkins

Building on the built-in node can be a security issue. You should set up distributed builds. See the documentation. Set up agent Set up cloud Dismiss

Warnings have been published for the following currently installed components: Go to plugin manager Configure which of these warnings are shown

Build Pipeline Plugin 2.0.2:
Stored XSS vulnerability (no fix available)
No fixes for these issues are available. It is recommended that you review the security advisory and apply mitigations if possible, or uninstall this plugin.

System Configuration

- System: Configure global settings and paths.
- Tools: Configure tools, their locations and automatic installers.
- Plugins: Add, remove, disable or enable plugins that can extend the functionality of Jenkins.
- Nodes: Add, remove, control and monitor the various nodes that Jenkins runs jobs on.
- Clouds: Add, remove, and configure cloud instances to provision agents on-demand.
- Appearance: Configure the look and feel of Jenkins.

Security

2 cm of rain Wednesday

LAB ACTION PLAN FOR WEEK x System - Jenkins Webhooks - Settings - Rohith6 Lab setup instructions App passwords

localhost:8080/manage/configure

Jenkins / Manage Jenkins / System

E-mail Notification

SMTP server: smtp.gmail.com

Default user e-mail suffix

Advanced Edited

☒ Use SMTP Authentication

User Name: kannarohith26255@gmail.com

For security when using authentication it is recommended to enable either TLS or SSL.

Password:

☒ Use SSL

☐ Use TLS

SMTP Port: 465

Reply-To Address: kannarohith26255@gmail.com

Save Apply

2 cm of rain Wednesday

LAB ACTION PLAN FOR WEEK x System - Jenkins x Webhooks - Settings - Rohith6 x Lab setup instructions x App passwords x +

localhost:8080/manage/configure

Jenkins Manage Jenkins / System

For security when using authentication it is recommended to enable either TLS or SASL

Password

Use SSL ☒ Use TLS ☐

SMTP Port 465

Reply-To Address kannarohith26255@gmail.com

Charset UTF-8

Test configuration by sending test e-mail

Test e-mail recipient kannarohith84999@gmail.com

Email was successfully sent

Test configuration

Save Apply

REST API Jenkins 2.516.3

2 cm of rain Wednesday

Search

ENG IN 12:02 PM 28-10-2025

SELAB_WEEK 11_1291 - G x System - Jenkins x Webhooks - Settings - Roh x Lab setup instructions x App passwords x mvnweb_build - Build # 14 x +

localhost:8080/manage/configure

Jenkins Manage Jenkins / System

SMTP server smtp.gmail.com

SMTP Port 465

Advanced Edited

Credentials kannarohith26255@gmail.com/***** + Add

Use SSL ☒ Use TLS ☐ Use OAuth 2.0 ☐

Advanced Email Properties

Save Apply

NIFTY -0.34%

Search

ENG IN 12:42 PM 28-10-2025

LAB ACTION PLAN FOR WEEK x System - Jenkins x Webhooks - Settings - Rohith6 x Lab setup instructions x App passwords x +

localhost:8080/manage/configure

Jenkins Manage Jenkins / System

Extended E-mail Notification

SMTP server
smtp.gmail.com

SMTP Port
465

Advanced Edited

Credentials
jenkins/***** + Add

☒ Use SSL
☐ Use TLS
☐ Use OAuth 2.0

Advanced Email Properties

Default user e-mail suffix

Save Apply

BAN - WI Game score Search ENG IN 12:04 PM 28-10-2025

LAB ACTION PLAN FOR WEEK x mvnweb_build Config - Jenkins x Webhooks - Settings - Rohith6 x Lab setup instructions x App passwords x +

localhost:8080/job/mvnweb_build/configure

Jenkins mvnweb_build / Configuration

Configure

General

Enabled

Description
web build demo

Plain text Preview

☐ Discard old builds
☐ GitHub project
☐ Permission to Copy Artifact
☐ This project is parameterised
☐ Throttle builds
☐ Execute concurrent builds if necessary

Advanced

Source Code Management

Connect and manage your code repository to automatically pull the latest code for your builds.

☐ None
☒ Git
Repositories

Save Apply

BAN - WI Game score Search ENG IN 12:05 PM 28-10-2025

The screenshot shows the Jenkins configuration page for the 'mmweb_build' job. The 'Editable Email Notification' section is expanded, showing the following settings:

- General:** Allows the user to disable the publisher, while maintaining the settings. ☐ Disable Extended Email Publisher.
- Project From:** (Empty text field)
- Project Recipient List:** Comma-separated list of email address that should receive notifications for this project.
kannarohith26255@gmail.com , kannarohith84999@gmail.com
- Project Reply-To List:** Comma-separated list of email address that should be in the Reply-To header for this project.
\$DEFAULT_REPLYTO
- Content Type:** Default Content Type (Dropdown menu)
- Default Subject:** \$DEFAULT_SUBJECT

At the bottom of the configuration page, there are 'Save' and 'Apply' buttons.

LAB ACTION PLAN FOR WEEK x mvnweb_build Config - Jenkins x Webhooks - Settings - Rohith60 x Lab setup instructions x App passwords x +

localhost:8080/job/mvnweb_build/configure

Configure

- General
- Source Code Management
- Triggers
- Environment
- Build Steps
- Post-build Actions

Attachments ?
Can use wildcards like 'module/dist/**/*.*'. See the [@includes of Ant fileset](#) for the exact format. The base directory is the [workspace](#).

Attach Build Log ?
Attach Build Log

Content Token Reference ?

Advanced Settings ^

Pre-send Script ?
\$DEFAULT_PRESEND_SCRIPT

Post-send Script ?
\$DEFAULT_POSTSEND_SCRIPT

Additional groovy classpath ?
Add

Save Apply

localhost:8080/job/mvnweb_build/configure#

NIFTY -0.38%

LAB ACTION PLAN FOR WEEK x mvnweb_build Config - Jenkins x Webhooks - Settings - Rohith60 x Lab setup instructions x App passwords x +

localhost:8080/job/mvnweb_build/configure

Configure

- General
- Source Code Management
- Triggers
- Environment
- Build Steps
- Post-build Actions

Success ?

Send To

Developers ?

Add

Advanced

Failure - 1st ?

Send To

Developers ?

Recipient List ?

Sends email to the list of recipients defined in the "Project Recipient List."

(from Email Extension Plugin)

Add

Save Apply

NIFTY -0.38%

ENG IN 12:09 PM 28-10-2025

```
[INFO] Processing war project
[INFO] Copying webapp resources [C:\ProgramData\jenkins\workspace\mnweb_build\src\main\webapp]
[INFO] Building war: C:\ProgramData\jenkins\workspace\mnweb_build\target\LibraryRegistration.war
[INFO]
[INFO] --- install:3.1.2:install (default-install) @ LibraryRegistration ---
[INFO] Installing C:\ProgramData\jenkins\workspace\mnweb_build\pom.xml to C:\WINDOWS\system32\config\systemprofile\.m2\repository\com\library\LibraryRegistration\1.0-SNAPSHOT\LibraryRegistration-1.0-SNAPSHOT.pom
[INFO] Installing C:\ProgramData\jenkins\workspace\mnweb_build\target\LibraryRegistration.war to C:\WINDOWS\system32\config\systemprofile\.m2\repository\com\library\LibraryRegistration\1.0-SNAPSHOT\LibraryRegistration-1.0-SNAPSHOT.war
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 3.550 s
[INFO] Finished at: 2025-10-28T12:41:07+05:30
[INFO] -----
Archiving artifacts
Email was triggered for: Always
Sending email for trigger: Always
Sending email to: kannarohith26255@gmail.com kannarohith84999@gmail.com
Triggering a new build of mnweb_test
Finished: SUCCESS
```

REST API Jenkins 2.516.3

The screenshot shows a Gmail interface on a Windows desktop. The browser tabs at the top include 'SELAB_WEEK 11_1291 - G...', 'mnweb_build #14 Console', 'Webhooks - Settings - Roh...', 'Lab setup instructions', 'App passwords', and 'mnweb_build - Build # 14...'. The Gmail inbox displays an email from 'mnweb_build - Build # 14 - Successful!' (Inbox x) sent 12:41 (0 minutes ago) to 'me, kannarohith84999'. The email body contains the text 'mnweb_build - Build # 14 - Successful:' and a link to check console output at 'http://localhost:8080/job/mnweb_build/14/'. An attachment 'build.log' is shown with a preview icon. The Windows taskbar at the bottom shows the NIFTY index at -0.34%, the search bar, and various application icons. The system clock indicates 12:41 PM on 28-10-2025.